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Evaluation of *Plasmodium berghei* for Endotoxin by the *Limulus* Lysate Assay

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The presence of endotoxin in the malaria parasite has long been suspected but has not been directly demonstrated by the *Limulus* amoebocyte lysate (LAL) assay or other assays. Tests for endotoxins have been done on plasma of infected individuals but not on the parasites themselves (Glew and Levin, 1975, Proc. Soc. Exp. Bio. Med. **148**: 508-510). Recently, Tubbs (1980, Trans. R. Soc. Trop. Med. Hyg. **74**: 121-123) detected endotoxin activity in the plasma of mice infected with *Plasmodium berghei* and in patients infected with *P. falciparum* using the LAL assay, and suggested that the endotoxin was derived from either the parasites or the endogenous bacteria in the gut. By assaying intact plasmodia washed free of contaminating host cells and plasma, it is possible to determine whether the plasmodia are the source of the endotoxin. We obtained free *P. berghei* using ultrasonic energy and tested the plasmodia for endotoxin activity using the LAL assay. The results obtained are the subject of this report.

Groups of mice and rats were infected by the intraperitoneal injection of approximately 10⁶ *P. berghei* parasites, and later the blood was collected aseptically by cardiac puncture when parasitemia reached 40%. The blood was transferred to pyrogen-free, sterile, heparinized tubes and washed three times in sterile, pyrogen-free 0.9% NaCl. Blood from uninfected mice was treated in an identical manner and served as a control. The blood specimens were then sonicated as previously

described (Prior and Kreier, 1972, Exp. Parasitol. **32**: 239-243). The continuous-flow chamber and sonicator probe tip were rendered pyrogen-free by heating at 160 C overnight. All equipment and solutions were tested for endotoxin contamination with the LAL test before use. Following sonication the specimens were centrifuged to harvest the plasmodia, and then adjusted to 25% packed-cell volume. The specimens were lysed by three cycles of freeze-thawing and assayed quantitatively for the presence of endotoxin using the microdilution LAL assay (Prior and Spagna, 1979, J. Clin. Microbiol. **10**: 394-395). The minimum sensitivity of the *Limulus* lysate utilized in the microdilution assay (courtesy of Mallinckrodt Inc., St. Louis, Missouri) was 0.06 ng/ml of LPS using the *E. coli* reference standard (lot EC-2, Bureau of Biologics, Food and Drug Administration). Aliquots of free parasite and control samples were seeded with known quantities of the *E. coli* endotoxin and assayed. Because certain heat-labile protein substances may give positive LAL test results, portions of the samples were heated at 100 C for 5 min and then tested for endotoxin activity. Parasites and control samples were also cultured on blood agar aerobically and anaerobically to detect any possible bacterial contamination.

The results obtained in four experiments using the LAL assay on the lysed parasite suspensions and controls are shown in Table I. A reaction suggestive of endotoxin was ob-

TABLE I. Results of LAL testing of free plasmodia and control preparations.

Samples	<i>Limulus</i> assay*				Mean ± SD†
	Exp. 1	2	3	4	
Free parasite suspension	1:4	<1:1	1:2	1:2	2.25 ± 1.26
Uninfected control	<1:1	<1:1	<1:1	<1:1	— —

* Highest dilution giving a positive LAL test.
† SD = standard deviation.

served in the parasite samples at dilutions of 1:4 or less, and this represents approximately 0.5 ng equivalent *E. coli* endotoxin per milliliter of packed parasites. This low activity was also present at the same dilutions following exposure to heat, suggesting that the positive LAL test results resulted from heat stable endotoxin. The samples seeded with *E. coli* endotoxin showed no reduction in activity, thus excluding the possibility that an inhibitor was present in our preparations. The undiluted (1:1) control samples gave negative LAL results. There was no bacterial growth on any of the cultures.

If *P. falciparum* malaria endotoxin activity were of the same order as *P. berghei* activity, then our preparation of greater than 10^{11} parasites/ml would be four log units greater than the mean number of parasites/ml in a human

with a 1% parasitemia (5.4×10^7 infected red cells/ml). Griesman and Hornick (1979, Proc. Soc. Exp. Biol. Med. **131**: 1154–1158) have shown that at least 1 ng of *E. coli* endotoxin/kg of body weight is needed to produce a fever in man. Therefore, if our results are applicable to the response in man, the endotoxin activity equivalent/kg of body weight in a man with a 1% parasitemia would be approximately three log units below the threshold level (1 ng/kg) required for fever induction.

From our studies, we conclude that the low endotoxin activity detected by the LAL assay in pure parasite preparations could not account for the fever seen in patients with malaria and that the endotoxin detected by others in plasma from malarious patients is probably derived from endogenous bacteria in the gut.

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Autocytotoxins During Rodent Malaria

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Autolymphocytotoxins have been observed in a number of human diseases including systemic lupus erythematosus (SLE), rheumatoid arthritis, scleroderma, Sjögren's syndrome, pernicious anemia, and multiple sclerosis, as well as following renal transplantation or certain immunizations (Pruzanski et al., 1978, Clin. Immunol. Immunopathol. **11**: 142–156).

Recently it has been suggested that autoantibodies to a subpopulation of T-lymphocytes are associated with active SLE and cause preferential depletion of cells involved with immune regulation (Twomey et al., 1978, J. Clin. Invest. **90**: 713–715). Thymocytotoxic autoantibody has also been observed in NZB mice, which develop autoimmune abnormalities similar to that seen in SLE in humans, and is capable of promoting in vitro phagocytosis of sensitized thymocytes by normal macrophages (Shirai and Mellors, 1971, Proc. Natl. Acad. Sci. **68**: 1412–1415). Recirculating

T-cells were found to be depleted selectively in NZB mice with progressive age (Denman and Denman, 1970, Clin. Exp. Immunol. **6**: 457–461). In a subsequent study the migratory pattern of labeled lymphocytes, pretreated with autoantibody, was followed in syngeneic mice (Shirai et al., 1973, J. Immunol. **110**: 517–523). The reported increase in the liver-localizing population lead the authors to suggest that T-cell depletion, with aging, is mediated by a continuous process of autosenescent phagocytosis. It was also observed that suppressor T-cells, and not helper T-cells, were sensitive to thymocytotoxic autoantibody (Shirai et al., 1978, J. Immunol. **120**: 1924–1929).

In the past we have observed a progressive depression in the splenic T-cell population during *P. berghei* infection in mice and an abnormal pattern of T-cell migration (Brisette et al., 1978, Proc. Soc. Exp. Biol. Med.